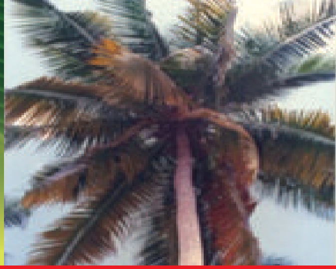
COCONUT RESEARCH INSTITUTE OF SRI LANKA

Advisory Circular No A 7



NUTRIENT DEFICIENCIES IN COCONUT PALMS

Most coconut growing soils in Sri Lanka are deficient in major plant nutrients which are Potassium (K), Magnesium (Mg), and Nitrogen (N). Nutrient deficiencies adversely affect the growth and production of coconut palms. Generally in acute mineral deficiencies characteristic symptoms could be detected on the foliage. Deficient nutrients could be identified at the field with the characteristic symptoms on the foliage. Sometimes mineral deficiencies can be present in palms without showing any visual symptoms. Such hidden deficiencies also adversely affect the nut production. Hidden mineral deficiencies can be identified only by chemical analysis of leaf.

Visual symptoms of deficiencies

The most commonly observed symptoms are due to potassium, magnesium, and nitrogen deficiencies. The occurrence of phosphorus deficiency is rare in coconut.

1. Nitrogen

Initial symptoms of nitrogen deficiency are light greenish color in the entire foliage. With the progress of nitrogen deficiency, fronts of the crown become yellow. (Water logging, severe drought, competition from grasses may also produce similar symptoms). Gradual reduction in the size of the crown and tapering of trunk could also be observed. In severe deficiencies, yellowing of fronds increases and turn to golden yellow and then to reddish gray before dying. Depending on the intensity of deficiency, the leaflets show uniform pale green to yellow colors (figure 1).

figure 1: Nitrogen deficiency showing palm

2. Potassium

Symptoms of potassium deficiency could initially be observed in mature fronds. Scattered rust- colored spots appear all over the leaflet. The leaflets become slightly yellowish and it is more prominent towards the tip. The reddish color spots gradually enlarged and from large brown patches. Leaflet tips show district scorching appearance. Fronts of deficient palm appear in yellow color while fronts in the lower world appear in orange yellow. Tapering of crown is a common feature of this deficiency (figure 2).

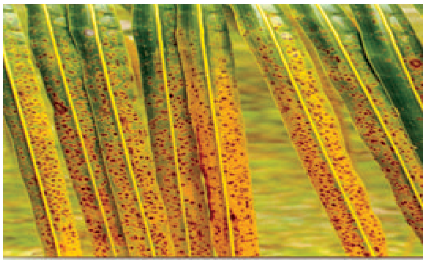


figure 2: Potassium deficiency showing coconut leaflets

figure 3: Potassium deficiency showing coconut palm

3. Magnesium

Magnesium deficiency is characterized by yellowing of leaflets of the mature fronds. The leaflets would be pale yellow with a green band on either sides of the ekel. The basal areas of leaflets remain green thereby showings a green band on either sides of the whole frond. Gradually leaflet tips dry, giving the appearance of the rachis of the whole frond. Gradually leaflet tips become dry and give scorching appearance (figure 3). At the mild stage of magnesium deficiency, the crown of the palm becomes pale yellow. When the intensity of magnesium deficiency gradually increases, yellowing of fronds in the lower world becomes prominent.

figure 4: Magnesium deficiency showing coconut leaflets

figure 5: Magnesium deficiency showing coconut palm

4. **Boron**

Due to Boron deficiency the leaflets of are wedded together and the tips of the leaflets appear like hooks. The lower basal region of the frond normally will not produce any leaflets (figure 4).

figure 6: Boron deficiency

Note: When symptoms are often due to deficiencies of more than one nutrient, it is difficult to conform exactly based on the above descriptions. Sometime deficiency symptoms mislead the growers due to temporary phenomenon as continuous drought or water logging. If growers are not in a position to diagnose the cause of the symptoms, they are advised to contact the nearest Coconut Development Officer (CDO) of the Coconut Cultivation Board (CCB).

Remedial measures

1. **Nitrogen deficiency**

Nitrogen deficiency occurs due to suspension of nitrogen fertilizer (urea, organic manure) application. Where fertilizer has been suspended for a short period, it is necessary to apply APM (adult palm mixture) or YPM (young palm mixture) fertilizer in the normal manner to correct the deficiencies caused by nitrogen (for additional information refer CRI advisory circular A5). If deficiency symptoms appear in spite of regular fertilizer application, apply 200g of urea per palm or 100 g of urea per young palm per year in addition to the application of APM or YPM fertilizers until the symptoms disappear.

2. **Potassium deficiency**

Potassium deficiency occurs due to suspension of potassium fertilizer (Muriate of Potash) application. If fertilizer application has been suspended for a short period, it is necessary to apply 'APM' fertilizer with dolomite in the normal manner to correct deficiencies caused by potassium (as recommended in leaflet No. A5). If deficiency symptoms appear in spite of regular APM fertilizer application, apply 500 g of Muriate of Potash per adult palm in addition to the application of recommended dose of APM fertilizer with dolomite.

3. Magnesium deficiency

❖ Old recommendation

If magnesium deficiency symptoms are visible apply Kieserite (24% MgO) as follows. The application should be continued until the deficiency is full recovered.

Young palms (1 year after planting up to bearing) – 0.5 kg per palm half yearly

Adult palms - 1kg per palm half yearly

Kieserite should be applied at least 3 months after application of fertilizer (APM or YPM fertilizer mixtures) when the soil is moist. If the deficiency is severe, suspend NPK fertilizer application for one year and apply only Kieserite as recommended above.

❖ New recommendation

In bearing palms with magnesium deficiencies, recommended NPK fertilizer (APM) is applied to the half of the manure circle and other half is applied with one kg of Kieserite at the same time. This application method should be continued until the palm is fully recovered from the deficiency. New recommendation minimizes the labor cost.

As a long-term preventive measure against magnesium deficiency, dolomite should be applied to adult coconut palms at the rate of 1 kg per palm per year along with NPK fertilizer application. For young palms 500g of dolomite is recommended at six-month intervals along with NPK fertilizers.

4. Boron deficiency

Boron deficiency symptoms could be observed in immature fronts of the bud region. In this deficiency the leaflets of the fronds fused together while showing zigzag nature (figure 4). Sometime on the base region of fronds leaflets are not formed. In the acute deficiency stage number of fronts in the crown reduces, bud region become black in color and stagnate the growth resulting the death of palm.

Corrective measures

By giving suitable treatments at the initial stage, the deficiency could be corrected. To correct boron deficiency applies 10 g of sodium tetra borate to coconut seedlings and 20 g for mature palms at 6 months interval until deficiency symptoms disappear. For fast recovery it is important to apply water to palm.

Soils and Plant Nutrition Division
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